

GANESH RAMESH

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SUMMARY

I am a serial Research Engineer, passionate about solving challenging problems requiring applications of Data Science, Artificial Intelligence and Machine Learning. I have over 15 years of experience in building and shipping products using large scale data as the backbone. I am currently exploring applications of Generative AI in agents and conversational assistants.

RESEARCH INTERESTS

Generative AI and agents, Large Scale applications of AI/ML, Search and Information Retrieval, Recommender Systems, Web/Text Mining and NLP, Product Data Science.

EDUCATION

Postdoctoral Fellow, Database Systems Lab

11/2003 – 10/2005

University of British Columbia

Supervisors: Dr. Laks V.S. Lakshmanan, Dr. Raymond Ng

Ph.D. Computer Science.

10/2003

University at Albany, State University of New York(SUNY)

Thesis: Data Mining Techniques for Frequent Itemsets: Construction and Analysis

Supervisor(s): Dr. William A. Maniatty, Dr. Daniel J. Rosenkrantz

M.S. (by Research) in Computer Science and Engineering:

06/1997

Indian Institute of Technology, Madras (Now Chennai)

Thesis: Design and Implementation of Tutoring Systems: An Intermediate Control Approach

B.E. Computer Science and Engineering

05/1994

University of Madras

Specialization: Parallel Processing

RESEARCH AND INDUSTRIAL EXPERIENCE

Samsung Research America

(Consulting) Gen AI Engineer

Mountain View, CA

November 2024 - Present

Application of Large Language Models and Generative AI technologies (RAG, fine tuning and prompt engineering) to the design and implementation of a health assistant for the Samsung Digital Health platform. Primary contributions are below:

1. Designed and shipped the memory module which stores short term and long term conversation history and related artifacts that provides relevant context to the assistant for grounding the responses to a user conversation.
2. Leading efforts to enhance debugging and visibility of various modules in the architecture as well as product and engineering metrics for measuring the effectiveness of the product.

3. Research on compact models and their applications to conversation summarization. Ongoing work on two papers related to small language models for summarization and related tasks.

Ema Unlimited
Research Engineer (Gen AI)

Mountain View, CA
December 2023 - August 2024

Machine learning lead for Generative AI technologies. My primary role is to develop the fundamental generative AI technology that will power downstream applications developed by Ema. Primary contributions are below:

1. Benchmarking Large Language Models for various types of tasks within the product ecosystem
2. Routing strategies for LLM calls to develop a low cost strategy without losses in accuracy
3. Developing robust design patterns for ensuring trace-ability of conversations within the product
4. Exploring smart composition strategies for response generation using LLMs.

Apple Inc.
Principal Research Engineer, AIML
Principal Research Engineer, Maps Search

Cupertino, CA
March 2022 - November 2023
August 2017 - February 2022

Key Achievements - Siri Query Understanding - Tech Lead

- An ML tool based on scientific principles that will provide insight about the quality of training data and help a user reason about model predictions. This tool was based on gradients and was extensively used for both spot fixing errors observed in prediction as well as for data cleaning.
- Fine tuned an address parser based on a BERT model which was shipped for the EN and JP markets.
- Fine tuned a parser model for app store search in collaboration with the app store team.

Key achievements - Maps Search - Architect/ML Lead for Address Ranker

1. Designed, implemented and productionized the end-to-end Machine Learning pipeline for building the address search ranker for various markets.
2. Replaced the manually built rule-based ranker which had become technically cumbersome to maintain.
3. Responsible for the project management including outbound communication as well as cross-team interaction for the project which spanned two years.
4. Shipped the ranker to 75% of address traffic by 2019 and the rest of the world by June 2020 serving hundreds of millions of queries worldwide across the entire Apple Device spectrum.
5. Led a cross team effort to enable the flow of probe data features which was one of the top 5 important features for the ranker model (Lifting the performance by 2 percentage points in observed online metrics)
6. Contributed to several architectural discussions and code/design reviews within the Maps team, including Query Tagging, Query Understanding, Feature engineering, Document Stores, Recall, Data Imputation, Machine Translation and Evaluation Metrics.

Management and Administration:

1. Mentored three junior team members in several aspects of Machine Learning, Search and Ranking. With my guidance and mentoring, I was able to transition the ownership of the ranker to them in Fall 2020 and focus on other projects.
2. Was an active mentor in the company mentorship program. Actively mentored 5 engineers across all functions within Apple.
3. Active organizer for the Apple ML summit. Was a part of the organization committee, the review committee and chaired the Knowledge Bases and Search Track for three consecutive years raising the bar for the quality of published and accepted work.

SAP

Principal Data Scientist

Palo Alto, CA
May 2016 - July 2017

1. Data exploration for intelligent procurement.
2. Enterprise search relevance and intelligent procurement agents.
3. Consulting Data Scientist for transforming personnel data and Data Privacy for Machine Learning.

Technologies/Languages: Java, Python, R, MySQL, MongoDB

Yahoo! Inc. (Yahoo! GEMINI)

Principal Research Engineer

Sunnyvale, CA
October 2015 - May 2016

1. Shipped two major improvements to sponsored search ad quality methods.
2. Using data driven analyses and metrics, worked with product managers, account managers (advertisers) and engineers to bridge the gap between advertiser expectations/happiness and the state of the ad serving system.
3. Designed data-driven metrics to engage with account managers to promote Yahoo sponsored search serving to advertisers.

Technologies/Languages: Java, Python, Pig, Hive, Hadoop

Edmodo Inc.

Principal Scientist/Engineer/Lead

San Mateo, CA
November 2013 - August 2015

1. Engineered a system to collect and curate educational content for recommendations. This system is a collection of general purpose and domain-specific code that curates and annotates resources extracted from the web for education.
2. Led a team that successfully shipped: a. A content recommendation service for Edmodo's assessment product. b. A content ecosystem that allows search and discovery of educational content complete with ratings and reviews of the products in the ecosystem. My contribution in code was in developing a significant part of the backend.
3. Designed and built a set of matching algorithms for educational objects. This allows for fast approximate match that powers many of our recommendations.

Technologies/Languages: Java, Python, Pig, Hive, Hadoop, MongoDB, MySQL

LinkedIn Corporation (Product Data Sciences)

Senior Research/Data Scientist

Mountain View, CA
April 2012 - November 2013

1. Designed and engineered the data pipeline for constructing session data for LinkedIn users based on their web and mobile page views.
2. Used features from session data to predict clickers (collaboration with my intern). The predictive model showed a 20% lift in predicting clickers when session based features were used in addition to existing features.
3. Engineered and shipped two location-based data products: a. Business traveler user segment which identified users who travel and b. Inferred location update which shipped as a part of the profile completion product.
4. Filed 4 patents in the areas of Inferred identity, recommendations and social reputation for professional social networks.
5. Engineered several data product prototypes: a. Gender inference for LinkedIn users. b. Recommending and Curating resources for learning skills. c. Aspirational, goal-oriented recommender system for LinkedIn users.

Technologies/Languages: Java, Python, Pig, Hive, R, Hadoop, MongoDB, MySQL

Yahoo! Labs (Advertising Sciences)

Senior Research Engineer

Santa Clara, CA

August 2009 - March 2012

1. Engineered parts of a state of the art supply forecasting system for large scale, complex multi-dimensional impression forecasting for Yahoo's Display Advertising platform.
2. Monitoring Tools and Metrics: Designed a comprehensive set of performance metrics for measuring stability, accuracy and consistency of forecasts for guaranteed display advertising contracts. Engineered the tools and data pipelines to compute these metrics.
3. Data Audits: Responsible for conducting several impactful data audits that validated the effectiveness of newly deployed serving algorithms and exposed several key issues leading to future improvements. Engineered the data pipeline and the tools to automate the performance of these audits.
4. Supply Landscape understanding: Applied pattern mining algorithms for identifying multidimensional regions that had a high sell through rate (areas of great demand for ads) and developed a methodology for navigating and presenting these regions. Also developed a method for identifying new regions with potential supply for those with high demand.

Technologies/Languages: Java, Python, Perl, Pig, Hive, Hadoop, MySQL

Microsoft Corporation(Windows Live Safety Platform), Antispam

Software Development Engineer

Redmond, WA

May 2007 - July 2009

1. **Mining Traceroute Data for Router Reputation:** Researched and engineered a scalable system that identifies and uses router statistics to filter spam e-mail. This system (in a conservative setting) helped reduce spam by 2%.
2. **Tools for Extracting Large Scale Network Data:** Engineered a multi-threaded generic network data collection tool for collecting network data (DNS, RDNS, Traceroute, Port scanners), significantly speeding up the throughput of data collection (upto 2 orders of magnitude in the best case) to baseline single threaded methods.
3. **Investigation of Natural Language Features for Classifying E-mail:** Studied the impact of generic natural language features extracted from graded mail to identify and classify spam e-mail.

4. **Tools for Automatic Extraction of Mail:** Engineered a tool for automatically extracting mails from various e-mail service providers (ESPs) and storing them into a structured relational database for spam statistics reporting.

Technologies/Languages: C#, SQL Server

Microsoft Corporation(Live Search)

Software Development Engineer

Redmond, WA

Jul 2006 - May 2007

1. **Mining and Monitoring Definitive Answers for Search Queries:** Definitive lists contain queries and their most relevant URLs in a cache for serving by the search engine, used as a failsafe mechanism. These lists are large, manually generated, multi-lingual, noisy and often ambiguous. I shipped language-insensitive methods based on click logs to identify potentially bad definitives. These techniques impacted upto 10% of the entries in many markets.
2. **Query Log Analysis for Information Vs Navigational Queries:** Performed a preliminary investigation of query and click logs to identify informational queries versus navigational queries - Queries for which people are seeking information spread across several URLs versus those that are satisfied by one single URL.

Technologies/Languages: C#, SQL Server

Amazon.com, Customer Behaviour Group

Analyst

Seattle, WA.

Oct 2005 - Jun 2006

1. **Optimizing Advertisement Content Delivery for Amazon Associates:** Designed, implemented and shipped a scalable machine learning model for recommending the best advertisements to be displayed on Amazon associate websites. Supported and further tuned the model to handle quality of recommendations and advanced phrasal features.

Technologies/Languages: Java, Perl, C++

Database Systems Lab, University of British Columbia

Postdoctoral Research Fellow

Vancouver, B.C.

Nov 2003 - Oct 2005

1. **Disclosure Risk Analysis of Anonymized Data:** We studied the problem of analyzing the risk in disclosing anonymized data to a third party. We considered the important case when the adversary or the third party has background knowledge about the data into our risk analysis. We proposed several practical methods using which a data owner can determine whether it is safe to release the data to the third party or not.
2. **Satisfiability of Tree Pattern Queries:** An important issue in evaluating XML queries is *satisfiability* i.e., given a query, can there exist a database on which the query has a non-empty answer (the database is expected to be consistent with respect to the schema, when one is present). We systematically studied satisfiability of tree pattern queries (which capture a useful fragment of XPath). We identified cases in which the problem can be solved in polynomial time and developed efficient algorithms for this purpose. We also identified several cases for which the problem is NP-complete. Our experience concludes through a systematic set of experiments that satisfiability check could effect substantial savings in query evaluation (Substantial savings for unsatisfiable queries and negligible overhead for satisfiable queries).

Avaya Labs Research

Research Intern

Basking Ridge, New Jersey

Aug 2001 - Nov 2001

Text Mining for Multimedia Clustering: Developed novel text mining techniques to learn patterns from a corpus of newswire articles. Developed a novel technique to filter commercial segments from closed caption

text. Using the mined results from newswire articles, we developed efficient methods to improve the sparsity of closed caption text. These methods were successfully applied in building a hybrid clustering algorithm prototype for indexing and retrieving broadcast news data. The hybrid clustering algorithm significantly improved the quality of clusters by linking several related news articles that were previously unidentified.

University at Albany, State University of New York
Research Assistant

Albany, New York
Sep 2000 - May 2003

1. **Distribution-Based Synthetic Database Generation:** Analyzed feasible itemset distributions in databases and developed novel strategies for producing synthetic databases for itemset mining.
2. **Indexing and Data Access Methods for Data Mining** - Conducted an extensive performance analysis of the impact of access methods and data layouts on the behaviour of representative itemset mining algorithms. Our goal was to integrate various data mining algorithms with existing data management technology by decoupling the algorithms from the actual storage technique.

Hewlett-Packard labs Research
Research Intern

Palo Alto, California
May 1999 - Aug 1999

Data Mining Applications: Applied decision tree and association rule based techniques to analyze HP data. Proposed and developed an association rule mining workbench to study and benchmark the performance of association rule mining algorithms.

General Electric Corporate Research and Development Labs
Research Intern

Schenectady, New York
May 1998 - Aug 1998

Customer Retention Models: Applied decision tree and regression based methods to develop models for retaining GE customers. This was successfully used to identify a list of potentially bad customers (based on which further action was taken).

TEACHING EXPERIENCE

University of British Columbia
Instructor: *Introduction to Relational Databases* - CPSC 304

Vancouver, B.C.
Term: Summer 2004.

University at Albany, SUNY
Instructor:

Systems Programming - CSI 402
Introduction to Discrete Mathematics - CSI 210

Albany, New York.
Terms: Summer 2000, 2002, 2003.
Term: Summer 2000.

Teaching Assistant:

Algorithms and Data Structures - CSI 403
Theory of Computation - CSI 409
Principles of Programming Languages - CSI 311
Programming the Hardware/Software Interface - CSI 333
Database Systems - CSI 508

Terms: Fall 1997, Fall 1999.
Terms: Spring 1998, Spring 1999.
Terms: Spring 2002, Spring 2003.
Term: Fall 1998.
Term: Spring 2000.

PROFESSIONAL SERVICE

Department of Computer Science, University at Albany, State University of New York

Member of Faculty Search Committee 1999-2000
Member of Faculty Search Committee 2000-2001

Organizing Committee Member

Simulation Day, University at Albany, SUNY, October 2002.

Discrete Mathematics and Computer Science Day (DMCSD), University at Albany, SUNY, September 2002.

Network Design Day, University at Albany, SUNY, October 1998.

Conference/Journal Referee

Program Committee Member: WIDM 2006 (Web Information and Data Management), KDD 2007 (Industrial track), DASFAA 2008, ICDM 2011 - 2013, 2015 - 2024, CIKM 2018, DBSocial workshop 2013, IEEE Big Knowledge 2017.

2009-2013: Reviewer for IEEE TKDE, ACM TKDD and VLDB Journal

2005: External reviewer for ICDE, VLDB, SIGMOD. Reviewer for IEEE Transactions on Knowledge and Data Engineering, DAMI.

2003, 2004: External Reviewer for SIGMOD, PAKDD, SIGKDD, VLDB and ICDM Conferences. Reviewer for IEEE Transactions on Knowledge and Data Engineering and VLDB Journal.

2002: External Reviewer for SIAM Conference on Data Mining.

AWARDS

University Rank Holder: Received a silver medal for obtaining the seventh highest overall GPA (out of over 1000 students) in my undergraduate program from the University of Madras.

TALKS AND PRESENTATIONS

1. **To Do or Not To Do: The Dilemma of Disclosing Anonymized Data**, at University of Victoria, July 2008, at ACM SIGMOD Conference, June 2005 and at the Colloquium in Honor of Dr. Daniel J. Rosenkrantz, University at Albany, May 6, 2005.
2. **On Testing Satisfiability of Tree Pattern Queries**, VLDB Conference, September 2004.
3. **The Combinatorics of Frequent Itemset Mining**, Database Systems Lab, University of British Columbia, June 2003.
4. **Feasible Itemset Distributions in Data Mining: Theory and Application**, ACM PODS Conference, June 2003.
5. **Indexing and Data Access Methods for Data Mining**, ACM SIGMOD Workshop on Data Mining and Knowledge Discovery (DMKD), June 2002.

PUBLICATIONS - Refereed Conferences and Workshops

1. Sean Chester, Bruce Kapron, Ganesh Ramesh, Gautam Srivastava, Alex Thomo and S. Venkatesh, **K-Anonymization of Social Networks by Vertex Addition**, in Proceedings II of 15th East-European Conference on Advances in Databases and Information Systems (ADBIS), ADBIS 2011 Research Communications, 2011.
2. Shaofeng Bu, Laks V.S. Lakshmanan, Raymond Ng and Ganesh Ramesh, **Preservation of Patterns and Input-Output Privacy**. In IEEE International Conference on Data Engineering **ICDE**, Istanbul, Turkey, 2007.
3. Ganesh Ramesh, **Can Attackers Learn from Samples?** In 2nd **VLDB** workshop on **Secure Data Management**, Trondheim, Norway, September 2005.
4. Ganesh Ramesh, Mohammed J. Zaki and William A. Maniatty, **Distribution-Based Synthetic Database Generation Techniques for Itemset Mining**. In 9th International Database Engineering and Applications Symposium **IDEAS**, Montreal, Canada, July 2005.

5. Laks V.S. Lakshmanan, Raymond Ng and Ganesh Ramesh, **To Do or Not To Do; The Dilemma of Disclosing Anonymized Data**. In **ACM SIGMOD** Conference, Baltimore, Maryland, June 2005.
6. Laks V.S. Lakshmanan, Ganesh Ramesh, Wendy (Hui) Hwang and Jessica (Zheng) Zhao, **On Testing Satisfiability of Tree Pattern Queries**. In **VLDB** Conference, Toronto, Canada, August 2004.
7. Ganesh Ramesh, William A. Maniatty and Mohammed J. Zaki, **Feasible Itemset Distributions in Data Mining: Theory and Application**. In **ACM PODS** Conference, San Diego, California, June 2003.
8. Ganesh Ramesh, William A. Maniatty and Mohammed J. Zaki, **Indexing and Data Access Methods for Database Mining**. In **ACM-SIGMOD Workshop on Data Mining and Knowledge Discovery (DMKD)**, Madison, Wisconsin, June 2002.
9. Amit Bagga, Jianying Hu, Jialin Zhong and Ganesh Ramesh, **Multi-source Combined-Media Video Tracking for Summarization**. In 16th International Conference on Pattern Recognition (**ICPR**), Quebec City, Canada, August 2002.
10. Ganesh Ramesh and Amit Bagga, **A Text-based Method for Detection and Filtering of Commercial Segments in Broadcast News**. In International Conference on Language Resources and Evaluation (**LREC**), Las Palmas, Canary Islands, Spain, May 2002.
11. Amit Bagga, Breck Baldwin and Ganesh Ramesh, **A Methodology for Cross-Document Coreference Over Degraded Data Sources**. In Recent Advances in Natural Language Processing (**RANLP**), Tzigrav Chark, Bulgaria, September 2001.

PUBLICATIONS - Refereed Journals:

1. Sean Chester, Bruce M. Kapron, Ganesh Ramesh, Gautam Srivastava, Alex Thomo, S. Venkatesh, **Why Waldo befriended the dummy? k-Anonymization of social networks with pseudo-nodes**. In Social Network Analysis and Mining, Volume 3, Number 3, pages 381-399, 2013.
2. Laks V.S. Lakshmanan, Raymond Ng and Ganesh Ramesh, **On Disclosure Risk Analysis of Anonymized Itemsets in the Presence of Prior Knowledge**. In ACM Transactions on Knowledge Discovery in Data (TKDD), Volume 2, Number 3, 2008.

PATENTS

1. **U.S. Patent No. 8812690 - August 19, 2014:** Method and system to provide reputation scores for a social network member. Inventors: **Ganesh Ramesh**, Ramakrishna Vemuri
2. **U.S. Patent No. 9635116 - April 25, 2017:** Techniques for inferring a location. Inventors: Sathyanarayan Anand, **Ganesh Ramesh**, Alexis Blevins Baird
3. **U.S. Patent No. 9661039 - May 23, 2017:** Recommending resources to members of a social network. Inventors: Heyning Cheng, Navneet Kapur, **Ganesh Ramesh**

Programming and Platforms

Languages: Java, Python, C#, C, C++, SQL

Frameworks and Cloud: Hadoop, Apache Spark, AWS, GCP

Employment Eligibility: United States Citizen.

REFERENCES: Available upon request.